$$= \vec{B} = \frac{\mu_0 I}{4\pi a}\left(\sin\phi_1 + \sin\phi_2\right)\hat{n}$$

For infinitely long conductor,

$\phi_1 = \phi_2 = 90°$

$$\therefore \vec{B} = \frac{\mu_0 I}{4\pi a} \times 2\,\hat{n} \Rightarrow \vec{B} = \frac{\mu_0 I}{2\pi a}\,\hat{n} \qquad (3.36)$$

3.8.3 Magnetic field produced along the axis of the current-carrying circular coil

Consider a current carrying circular loop of radius R and let I be the current flowing through the wire in the direction as shown in Figure 3.33.

The magnetic field at a point P on the axis of the circular coil at a distance z from the centre of the coil O is computed by taking two diametrically opposite line elements of the coil each of length $d\vec{l}$ at C and D. Let $\vec{r}$ be the vector joining the current element ($I\,d\vec{l}$) at C and the point P.

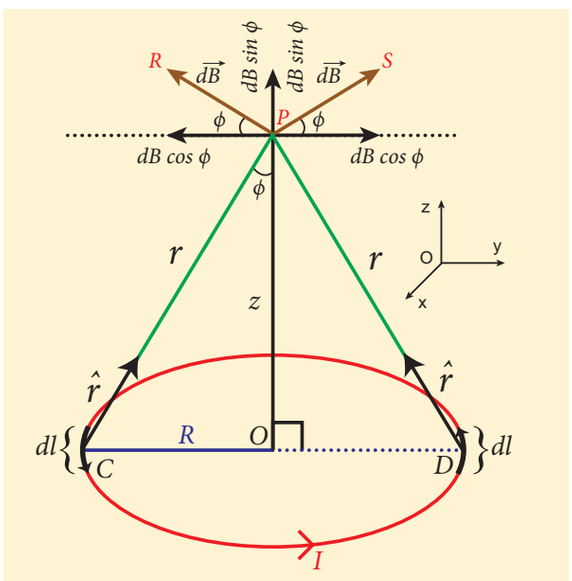


Figure 3.33 Magnetic field due to current-carrying circular loop

According to Biot-Savart's law, the magnetic field at P due to the current element at C is

$$d\vec{B} = \frac{\mu_\circ}{4\pi}\frac{I\,d\vec{l} \times \hat{r}}{r^2}$$

The magnitude of $d\vec{B}$ is

$$dB = \frac{\mu_\circ}{4\pi}\frac{I\,dl\sin\theta}{r^2} = \frac{\mu_\circ}{4\pi}\frac{I\,dl}{r^2}$$

where θ is the angle between $Id\vec{l}$ and $\vec{r}$. Here $\theta = 90°$.

The direction of $d\vec{B}$ is perpendicular to the current element $Id\vec{l}$ and CP. It is therefore along PR perpendicular to CP.

The magnitude of magnetic field at P due to current element at D is same as that for the element at C because of equal distances from the coil. But its direction is along PS.

The magnetic field $d\vec{B}$ due to each current element is resolved into two components; $dB\cos\phi$ along y-direction and $dB\sin\phi$ along z-direction. The horizontal components cancel out while the vertical components ($dB\sin\phi\,\hat{k}$) alone contribute to the net magnetic field $\vec{B}$ at the point P.

$$\vec{B} = \int d\vec{B} = \int dB\sin\phi\,\hat{k}$$

$$= \frac{\mu_0 I}{4\pi}\int \frac{dl}{r^2}\sin\phi\,\hat{k}$$

From ΔOCP,

$$\sin\phi = \frac{R}{\left(R^2 + z^2\right)^{1/2}} \text{ and } r^2 = R^2 + z^2.$$

Substituting these in the above equation, we get

$$\vec{B} = \frac{\mu_0 I}{4\pi}\frac{R}{\left(R^2 + z^2\right)^{3/2}}\hat{k}\left(\int dl\right)$$

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